

Scalability & Future Plan

Date	2 October 2026
Team ID	SWTID-2026-3138
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Image generation with Stable Diffusion is slow and needs a powerful GPU	Long waiting time for users, and fewer comics can be created at the same time	High
2	Character appearance can change between panels	Comic looks less consistent and story quality is reduced	High
3	Dependence on Gemini API limits and internet connection	Story generation may fail or slow down when usage is high or the network is poor	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI server handling requests one by one	Deploy on cloud with load balancing and a task queue (e.g., Celery / Redis) to handle many users together
2	Data Storage	Generated stories and images are stored locally	Move to cloud storage (e.g., AWS S3 / Google Cloud Storage) and a database for user accounts and saved comics
3	Performance	Images generated one panel at a time	Use GPU instances, batch image generation and caching of repeated prompts
4	Security	Basic API key protection	Add user login, secure storage of API keys, rate limiting and content safety filters

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Character consistency across panels and more art styles (manga, cartoon, realistic)	1 - 2 months	Better quality and more choice for users
Phase 2	User accounts, saved comics gallery and sharing option	3 - 4 months	Users can manage and share their comics
Phase 3	Mobile app, multi-language stories and voice-based story input	6 months	Wider reach and easier access for more users